Proofs of the Kozma and Nitzan Gluing Conjectures for Bernoulli Percolation

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Abstract

Kozma and Nitzan conjectured inequalities comparing the probability that one vertex of a finite weighted graph is joined to another with the probabilities that a prescribed finite set of vertices is joined to each of them, and they proved that the weakest of these implies that critical Bernoulli bond percolation on the hypercubic lattice has no infinite open cluster in every dimension at least two. Their paper contains nine numbered statements. This note proves eight of them and records the exact status of the ninth. The whole argument rests on one inequality, which bounds the mean of an increasing function of the open cluster of a vertex from below by the mean of the same function at the first vertex of the prescribed set that the cluster reaches, and which is proved by removing the vertices of that set one at a time. Two features of it carry all the strength: the whole inequality may be conditioned on the cluster missing an auxiliary set of vertices, and a set of two vertices may replace the single starting vertex, which is what turns it into a statement about a graph with one edge forced open.

1 Introduction

Let every unordered pair of vertices of a finite vertex set be open independently with a prescribed probability, and write $x \leftrightarrow y$ when the two vertices are joined by a path of open pairs. Kozma and Nitzan asked how the probability that a vertex o is joined to a vertex b compares with the probabilities that o reaches a prescribed finite set A and that the vertices of A reach b. A path from o to A and a path from a vertex of A to b should combine into a path from o to b, but the two paths are not independent, and no proof of the resulting inequality was known. Their paper states five numbered conjectures and four numbered questions of this shape and proves that the weakest of the conjectures implies that critical Bernoulli bond percolation on the hypercubic lattice of dimension at least two has no infinite open cluster.

Eight of the nine numbered statements are proved here. Conjectures 1, 2, 3 and 4 and Questions 5 and 7 are theorems, in forms stronger than the printed ones. Conjecture 6 is a theorem, without either of the two hypotheses printed with it. Question 9 is answered affirmatively. Question 8 has an affirmative answer if a tie among the candidate vertices is excluded, and its printed reading, in which every candidate is allowed, is false: an example with four vertices is given in Section 9.

Everything rests on one inequality. Order the vertices of a finite set T by the mean of an increasing function F of the open cluster, and let J_x be the event that x is the first vertex of T that the open cluster of o reaches. The inequality asserts

$$\sum_{x \in T} \mathbb{P}(J_x)\, \mathbb{E}F(C_x) \;\le\; \mathbb{E}\big[F(C_o);\, o \leftrightarrow T\big] \, :$$

the mean of F at o, restricted to the event that o reaches T, is at least the mean of F at the vertex through which T is first reached. For the indicator of the event that b lies in the cluster this is a statement about connection probabilities, and it is the inequality from which Conjecture 3 was derived.

Two extensions of it carry the present note. The first conditions every expectation on the open cluster missing a prescribed set Y of vertices, Theorem 4.1. Applied with Y the single vertex a at which the mean of F is smallest, and applied not to F but to the function Φ_a of Section 5, which subtracts from $F(K)$ the mean of F at a in a fresh configuration with all pairs meeting K deleted, it yields Conjecture 4 with the vertex a prescribed in advance, Theorem 5.2, and with it Conjectures 1, 2 and 3 and Questions 5 and 7.

The second extension replaces the single vertex o by a set of two vertices and the cluster C_o by the union of the two clusters, Theorem 7.16. That union, and the events that it misses a set and that it meets a set, do not depend on the state of the pair joining the two vertices. The inequality therefore holds verbatim in the graph in which that pair is forced open, and Conjecture 6 follows. Question 9 follows from the same theorem with a random set of two or more vertices, obtained by exposing the pairs at o.

Sections 2 and 3 fix the model, state the nine numbered items in the vocabulary of the paper, and record the inputs. Section 4 proves Theorem 4.1, and Sections 5 and 6 deduce Conjectures 1 to 4 and Questions 5 and 7. Section 7 treats Conjecture 6, Section 8 treats Question 9 and Section 9 treats Question 8. Section 10 records for each statement how it has been checked.

2 The model and the nine numbered statements

Let V be a finite set and let every unordered pair e of elements of V be open with probability $p_e \in [0,1]$, independently over the pairs. A pair of weight zero is an absent edge, so this is percolation on a finite weighted graph with vertex set V. Write $\mathbb{P}$ and $\mathbb{E}$ for the resulting probability and expectation. For vertices x and y, write $x \leftrightarrow y$ when there is a path of open pairs from x to y, and $x \not\leftrightarrow y$ otherwise; connection is reflexive, so $x \leftrightarrow x$ always. Write

$$C_x = \{y \in V :\ x \leftrightarrow y\}$$

for the open cluster of x and $\mathcal{C}_x$ for the set of open pairs having at least one endpoint in C_x. For a set B of vertices, the event $\{x \leftrightarrow B\}$ is $\{C_x \cap B \neq \varnothing\}$ and $\{x \not\leftrightarrow B\}$ is its complement. A function F from subsets of V to the real numbers is increasing if $F(K) \le F(L)$ whenever $K \subseteq L$. Every set in this note is finite, and A is always a nonempty subset of V.

Kozma and Nitzan write $\mathbb{P}(0 \leftrightarrow b)$ for what is written $\mathbb{P}(o \leftrightarrow b)$ here, and they call a function f of a vertex and a configuration a monotone cluster property if $f(v,\omega)$ depends only on the open cluster of v and is increasing in it, and if $f(v,\omega) = f(w,\omega)$ whenever the pair $\{v,w\}$ is open. Their five conjectures and four questions are the following. There is no Conjecture 5 and no Question 6.

Conjecture 2.1 (Kozma and Nitzan, display (1))**.** Let o and b be vertices and let A be a nonempty set of vertices. Then

$$\mathbb{P}(o \leftrightarrow b)\ \ge\ \mathbb{P}(o \leftrightarrow A)\ \min_{x\in A} \mathbb{P}(x \leftrightarrow b).$$

Conjecture 2.2 (Kozma and Nitzan, display (2))**.** Let o and b be vertices and let A be a nonempty set of vertices. Then

$$\mathbb{P}(o \leftrightarrow b)\ \ge\ \min_{x\in A} \mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b).$$

Kozma and Nitzan call Conjecture 2.1 the post-FKG conjecture and Conjecture 2.2 the pre-FKG conjecture, and they record the formally stronger form

$$\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \ \geq\ \min_{x \in A} \mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b), \tag{1}$$

their display (3).

Conjecture 2.3 (Kozma and Nitzan, page 15)**.** For every $\varepsilon > 0$ there is a $\delta > 0$ such that, for every finite weighted graph, every nonempty set A of vertices and all vertices o and b with $\mathbb{P}(o \leftrightarrow A) > 1 - \delta$ and with $\mathbb{P}(x \leftrightarrow b) > 1 - \delta$ for every $x \in A$, one has $\mathbb{P}(o \leftrightarrow b) > 1 - \varepsilon$.

Conjecture 2.4 (Kozma and Nitzan, page 32)**.** Let f be a monotone cluster property, let A be a nonempty set of vertices and let o be a vertex. Then

$$\mathbb{E}\big[f(o, \cdot)\,;\, o \leftrightarrow A\big] \ \geq\ \min_{x \in A} \mathbb{E}\big[f(x, \cdot)\,;\, o \leftrightarrow A\big].$$

For a pair e, write $\mathbb{P}^e$ for the percolation measure in which the weight of e is replaced by one and all other weights are unchanged. Kozma and Nitzan write $G \downarrow e$ for the corresponding graph, which is the contraction of e as far as connection events are concerned.

Conjecture 2.5 (Kozma and Nitzan, displays (39) and (40))**.** Let A be a nonempty set of vertices, let b be a vertex, let $a \in A$ minimise $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$, and let $e = \{v, w\}$ be a pair. Assume

$$\mathbb{P}(x \leftrightarrow b) \ \geq\ \mathbb{P}(x \leftrightarrow A)\,\mathbb{P}(a \leftrightarrow b) \qquad \text{for } x = v \text{ and } x = w. \tag{2}$$

Then

$$\mathbb{P}^e(v \leftrightarrow b) \ \geq\ \mathbb{P}^e(v \leftrightarrow A)\,\mathbb{P}^e(a \leftrightarrow b). \tag{3}$$

Question 2.6 (Kozma and Nitzan, display (38))**.** Let A be a nonempty set of vertices and let o be a vertex. Are there numbers $c_x \geq 0$ for $x \in A$ with $\sum_{x \in A} c_x = 1$, not depending on b, such that

$$\mathbb{P}(o \leftrightarrow b) \ \geq\ \sum_{x \in A} c_x\, \mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b) \qquad \text{for every vertex } b?$$

The last three questions all ask whether a specified vertex $a \in A$ satisfies the inequality of the paper's display (41),

$$\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \ \geq\ \mathbb{P}(o \leftrightarrow A,\ a \leftrightarrow b). \tag{4}$$

Question 2.7 (Kozma and Nitzan, page 36)**.** Does (4) hold when a minimises $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$?

Question 2.8 (Kozma and Nitzan, page 36)**.** Does (4) hold when a minimises $\mathbb{P}(x \leftrightarrow b,\ o \not\leftrightarrow A)$ over $x \in A$?

Question 2.9 (Kozma and Nitzan, page 36)**.** Let H be the graph obtained from the given one by deleting every pair incident to o. Does (4) hold, with all probabilities evaluated in the original graph, when a minimises $\mathbb{P}_H(x \leftrightarrow b)$ over $x \in A$?

The minimand in Question 2.8 is the probability of the joint event that x reaches b and that o does not reach A; it is not the minimand of (4) itself, which is treated by (1).

3 The two inputs

Two families of inequalities are taken as given. The first is a conditional positive association inequality for one open cluster with two prescribed sets that the cluster must miss, proved by van den Berg, Häggström and Kahn.

Theorem 3.1 (van den Berg, Häggström and Kahn)**.** *Let s be a vertex, let X_1 and X_2 be sets of vertices not containing s and let g_1 and g_2 be nonnegative increasing functions of sets of pairs. Then*

$$\mathbb{E}\big[g_1(\mathcal{C}_s);\, s \not\leftrightarrow X_1\big]\,\mathbb{E}\big[g_2(\mathcal{C}_s);\, s \not\leftrightarrow X_2\big] \;\le\; \mathbb{E}\big[g_1(\mathcal{C}_s)g_2(\mathcal{C}_s);\, s \not\leftrightarrow X_1 \cap X_2\big]\,\mathbb{P}(s \not\leftrightarrow X_1 \cup X_2).$$

Moreover, if S and T are disjoint sets of vertices, if g is increasing in the set of open pairs met by paths from S and if h is increasing in the set of open pairs met by paths from T, then, with $D = \{S \not\leftrightarrow T\}$,

$$\mathbb{P}(D)\,\mathbb{E}[g\,h;\, D] \;\le\; \mathbb{E}[g;\, D]\,\mathbb{E}[h;\, D].$$

Taking $X_1 = X_2$ in the first display gives the conditional positive association of two increasing functions of a single cluster,

$$\mathbb{E}\big[g_1(\mathcal{C}_s);\, s \not\leftrightarrow X\big]\,\mathbb{E}\big[g_2(\mathcal{C}_s);\, s \not\leftrightarrow X\big] \;\le\; \mathbb{P}(s \not\leftrightarrow X)\,\mathbb{E}\big[g_1(\mathcal{C}_s)g_2(\mathcal{C}_s);\, s \not\leftrightarrow X\big], \tag{5}$$

and taking $g_1 = g_2 = 1$ in the first display gives the Harris inequality for the two decreasing events $\{s \not\leftrightarrow X_1\}$ and $\{s \not\leftrightarrow X_2\}$. The Harris inequality for two increasing events is used repeatedly and without further comment. The Ahlswede and Daykin four functions theorem is used once, in Lemma 7.11.

The second input is a family of covariance inequalities for one open cluster, conditioned on that cluster missing a prescribed set of vertices, and discounted by a finite list of auxiliary vertices. Fix a vertex x, a set Y of vertices with $x \notin Y$, a list $z_1, \ldots, z_\ell$ of distinct vertices and two further distinct vertices o and v, with all the named vertices outside Y and pairwise distinct. Write ν for the conditional law $\mathbb{P}(\,\cdot \mid x \not\leftrightarrow Y)$ and put

$$c_j(u) = \mathbb{P}\big(u \in C_{z_j} \,\big|\, z_j \not\leftrightarrow Y_j\big), \qquad \tau = \mathbb{P}\big(o \in C_v \,\big|\, v \not\leftrightarrow Y_{\ell+1}\big), \tag{6}$$

where $Y_j = \{x\} \cup Y \cup \{z_1, \ldots, z_{j-1}\}$ for $1 \le j \le \ell + 1$. For a real function φ on V put $\varphi^0 = \varphi$ and

$$\varphi^j(u) = \varphi^{j-1}(u) - c_j(u)\,\varphi^{j-1}(z_j) \quad (1 \le j \le \ell), \qquad \mathrm{M}[\varphi] = \varphi^\ell(o) - \tau\,\varphi^\ell(v). \tag{7}$$

The jth step removes from the value at every vertex the part explained by z_j, at the rate at which z_j reaches that vertex while missing everything named before it. All the constants are computed under $\mathbb{P}$ and not under ν.

Theorem 3.2. *Let x, Y, the list $z_1, \ldots, z_\ell$ and the vertices o and v be as above, let every pair weight lie strictly between zero and one, and let f be an increasing function of sets of pairs. Then*

$$\mathrm{M}\big[\,u \mapsto \mathrm{Cov}_\nu\big(f(\mathcal{C}_x), \mathbf{1}\{u \in C_x\}\big)\,\big] \;\ge\; 0.$$

For $\ell = 0$ the conclusion reads

$$\mathrm{Cov}_\nu\big(f(\mathcal{C}_x), \mathbf{1}\{o \in C_x\}\big) \;\ge\; \mathbb{P}\big(o \in C_v \,\big|\, v \not\leftrightarrow \{x\} \cup Y\big)\; \mathrm{Cov}_\nu\big(f(\mathcal{C}_x), \mathbf{1}\{v \in C_x\}\big),$$

and for $\ell = 0$, $Y = \varnothing$ and $f = \mathbf{1}\{v \in \,\cdot\,\}$ it is the Harris inequality.

Theorem 3.2 also carries with it a decomposition that will be used in Section 7, and which is recorded here because it is a property of the objects appearing in its proof rather than of its statement. Condition on the set $\mathcal{C}_Y$ of open pairs met by paths from Y. On the event $D = \{x \not\leftrightarrow Y\}$, the vertex set $C_Y = Y \cup \operatorname{span} \mathcal{C}_Y$ misses C_x, every pair with exactly one endpoint in C_Y is closed, and the pairs with both endpoints outside C_Y retain their original independent weights. Consequently, writing $\bar{f}$ for the mean of $f(\mathcal{C}_x)$ in a fresh configuration in which every pair meeting C_Y is deleted, the random variable

$$\Theta = \mathbf{1}_D\big(f(\mathcal{C}_x) - \bar{f}\,\big) \tag{8}$$

has $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$. For a set K of vertices disjoint from $\{x\} \cup Y$, let $\Psi(K)$ be minus the mean of Θ computed in the configuration in which every pair meeting K is deleted. Then $\Psi(\varnothing) = 0$, the function Ψ is nonnegative and increasing, and for every vertex z and every set B containing $\{x\} \cup Y$ but not z,

$$\mathbb{E}\big[\Theta;\, z \not\leftrightarrow B\big] \;=\; -\,\mathbb{E}\big[\Psi(C_z);\, z \not\leftrightarrow B\big]. \tag{9}$$

Both the vanishing of the conditional mean of Θ and the identity (9) express the domain Markov property of percolation: given the complete set of open pairs met by paths from a set, the configuration outside the cluster of that set is fresh.

We turn to the inequality that all of the numbered statements are deduced from.

4 A lower bound conditioned on missing a fixed set

Fix a set Y of vertices and an increasing function F of vertex sets. For a vertex x with $\mathbb{P}(x \not\leftrightarrow Y) > 0$ put

$$m_x = \frac{\mathbb{E}\big[F(C_x);\, x \not\leftrightarrow Y\big]}{\mathbb{P}(x \not\leftrightarrow Y)}, \tag{10}$$

the mean of F at x given that x misses Y. Let T be a finite set of vertices disjoint from Y with $\mathbb{P}(x \not\leftrightarrow Y) > 0$ for every $x \in T$, and let r be an injective real function on T such that $m_x \le m_y$ whenever $r(x) < r(y)$. Such a function exists: order T by the values m_x and break ties arbitrarily. For $x \in T$ and a vertex o write

$$J_x = \{o \not\leftrightarrow Y\} \cap \{o \leftrightarrow x\} \cap \bigcap_{y \in T,\ r(y) < r(x)} \{o \not\leftrightarrow y\} \tag{11}$$

for the event that o misses Y and that x is the vertex of smallest value of r that the open cluster of o reaches in T. The events J_x for $x \in T$ are disjoint and their union is $\{o \not\leftrightarrow Y\} \cap \{o \leftrightarrow T\}$. Put

$$\Lambda_o(T) = \mathbb{E}\big[F(C_o);\, o \not\leftrightarrow Y,\ o \leftrightarrow T\big] - \sum_{x \in T} \mathbb{P}(J_x)\, m_x. \tag{12}$$

Theorem 4.1. *Let V be a finite vertex set with pair weights in $[0,1]$, let $Y \subseteq V$, let F be an increasing function of subsets of V, let $T \subseteq V \setminus Y$ satisfy $\mathbb{P}(x \not\leftrightarrow Y) > 0$ for every $x \in T$, let r be injective on T with $m_x \le m_y$ whenever $r(x) < r(y)$, and let $o \in V$. Then $\Lambda_o(T) \ge 0$, that is*

$$\sum_{x \in T} \mathbb{P}(J_x)\, m_x \;\le\; \mathbb{E}\big[F(C_o);\, o \not\leftrightarrow Y,\ o \leftrightarrow T\big].$$

For $Y = \varnothing$ this is the inequality quoted in the introduction. No sign hypothesis on F is needed. The proof removes from T the vertex at which m is largest and transfers the resulting deficit from that vertex back to o. The transfer step is the following inequality, which is what Theorem 3.2 supplies.

Proposition 4.2. *Let V, Y, F, T and r be as in Theorem 4.1, and let $v \in V$ lie outside $Y \cup T$. Then, for every vertex o,*

$$\mathbb{P}\big(v \nleftrightarrow Y \cup T,\ o \leftrightarrow v\big)\, \Lambda_v(T) \ \le\ \mathbb{P}\big(v \nleftrightarrow Y \cup T\big)\, \Lambda_o(T).$$

Proof. Assume first that every pair weight lies strictly between zero and one and that o lies outside $Y \cup T$ and differs from v; the remaining positions of o are treated at the end. Let $z_1, \dots, z_\ell$ be distinct vertices outside $Y \cup T \cup \{o, v\}$ and let M be the functional (7) built from the constants (6) with Y replaced by $Y \cup T$ at every step, so that z_j is conditioned to miss $Y \cup T \cup \{z_1, \dots, z_{j-1}\}$ and the final constant is $\tau = \mathbb{P}(o \in C_v \mid v \nleftrightarrow Y \cup T \cup \{z_1, \dots, z_\ell\})$. We show by induction on the cardinality of T, simultaneously for every such list and every pair of vertices o and v, that

$$\mathrm{M}\big[\, u \mapsto \Lambda_u(T) \,\big] \ \ge\ 0. \tag{13}$$

For $T = \varnothing$ each $\Lambda_u(\varnothing)$ vanishes. Let T be nonempty, let $k \in T$ have the largest value of r and put $T_0 = T \setminus \{k\}$. Write $E = \{k \nleftrightarrow Y \cup T_0\}$.

We first record an identity. Since k has the largest value of r, adding k to T_0 leaves the events J_x for $x \in T_0$ unchanged and creates the single new event $\{u \nleftrightarrow Y\} \cap \{u \leftrightarrow k\} \cap \{u \nleftrightarrow T_0\}$, which equals $E \cap \{u \leftrightarrow k\}$: indeed on $\{u \leftrightarrow k\}$ the clusters of u and k coincide. On that event $F(C_u) = F(C_k)$, so

$$\Lambda_u(T) = \Lambda_u(T_0) + \mathbb{E}\big[F(C_k);\, E \cap \{u \leftrightarrow k\}\big] - \mathbb{P}\big(E \cap \{u \leftrightarrow k\}\big)\, m_k. \tag{14}$$

The three terms on the right are, in turn, a function of u whose M value is nonnegative by the induction hypothesis, a multiple of the conditioned covariance appearing in Theorem 3.2 with owner k, and a multiple of the probability that k reaches u while missing $Y \cup T_0$. Precisely, with $\gamma_k = m_k \mathbb{P}(E) - \mathbb{E}[F(C_k); E]$,

$$\mathbb{P}(E)\Big(\mathbb{E}\big[F(C_k); E \cap \{u \leftrightarrow k\}\big] - \mathbb{P}\big(E \cap \{u \leftrightarrow k\}\big) m_k\Big) = \mathrm{Cov}^*(u) + \gamma_k\, \mathbb{P}\big(E \cap \{u \leftrightarrow k\}\big), \tag{15}$$

where $\mathrm{Cov}^*(u) = \mathbb{P}(E)\mathbb{E}[F(C_k); E \cap \{u \leftrightarrow k\}] - \mathbb{E}[F(C_k); E]\mathbb{P}(E \cap \{u \leftrightarrow k\})$ is the conditioned covariance of Theorem 3.2, written without denominators, for the owner k, the set $Y \cup T_0$ and the increasing function $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$. The identity (15) is obtained by expanding both sides. Theorem 3.2 for the owner k, the set $Y \cup T_0$, the same list and the same two vertices gives $\mathrm{M}[\mathrm{Cov}^*] \ge 0$. Theorem 3.2 for the same owner and set and for the increasing function $\mathcal{C} \mapsto \mathbf{1}\{\mathcal{C} \neq \varnothing\}$ gives $\mathrm{M}[u \mapsto \mathbb{P}(E \cap \{u \leftrightarrow k\})] \ge 0$: on $\{u \leftrightarrow k\}$ with $u \neq k$ the cluster of k contains an open pair, and on E its mean is $\mathbb{P}(E)$ minus the probability that k misses $Y \cup T_0$ and is isolated, so the covariance in question is that isolation probability times $\mathbb{P}(E \cap \{u \leftrightarrow k\})$, and the isolation probability is positive because all weights are strictly below one. Finally,

$$\gamma_k \ \le\ \Lambda_k(T_0). \tag{16}$$

Indeed the definition of m_k gives $\mathbb{E}[(F(C_k) - m_k); k \nleftrightarrow Y] = 0$; splitting $\{k \nleftrightarrow Y\}$ into E and $\{k \nleftrightarrow Y\} \cap \{k \leftrightarrow T_0\}$ turns this into $\gamma_k = \mathbb{E}[(F(C_k) - m_k); k \nleftrightarrow Y, k \leftrightarrow T_0]$, and expanding the latter over the disjoint events J_x of (11) taken with k in place of o gives $\gamma_k = \Lambda_k(T_0) + \sum_{x \in T_0} \mathbb{P}(J_x)(m_x - m_k)$, in which every summand is at most zero because $r(x) < r(k)$. Applying M to (14), using (15), the two consequences of Theorem 3.2, (16) and the induction hypothesis for T_0 with the list extended by k, and dividing by $\mathbb{P}(E) > 0$, gives (13).

Take now $\ell = 0$ in (13). The conclusion is $\Lambda_o(T) \ge \tau \Lambda_v(T)$ with $\tau = \mathbb{P}(o \in C_v \mid v \nleftrightarrow Y \cup T)$, which is the assertion after multiplication by $\mathbb{P}(v \nleftrightarrow Y \cup T) > 0$. If $o = v$ the assertion is trivial; if

$o \in Y$ then $\Lambda_o(T) = 0$ and $\mathbb{P}(o \nleftrightarrow Y) = 0$ forces the left side to vanish; if $o \in T$ then $\Lambda_o(T) \geq 0$ directly from (12), since on J_o the value $F(C_o)$ is at least m_o and J_x is null for x with $r(x) > r(o)$.

It remains to pass from weights strictly inside the unit interval to weights in $[0, 1]$. When r is compatible with the numbers m_x as above, the quantity (12) admits the expression

$$\Lambda_u(T) = \mathbb{E}\Big[\mathbf{1}\{u \nleftrightarrow Y,\, u \leftrightarrow T\}\Big(F(C_u) - \min_{x \in T \cap C_u} m_x\Big)\Big], \tag{17}$$

which does not refer to r: on J_x the vertex x has the smallest value of r among those of T in C_u, hence by compatibility the smallest value of m. Every event probability is a polynomial in the pair weights, and each m_x is a ratio of two polynomials whose denominator does not vanish at the given weights, so both sides of the asserted inequality are continuous at the given weight vector. Approximating that vector by weight vectors with all coordinates strictly inside the unit interval, choosing for each of them an injective compatible r, and passing to a subsequence along which the induced order on T is constant, the inequality holds for the approximating vectors by the case already proved together with (17), and therefore at the given vector. □

Proof of Theorem 4.1. Induct on the cardinality of T. For $T = \varnothing$ the quantity (12) vanishes. Let T be nonempty, let $k \in T$ have the largest value of r, put $T_0 = T \setminus \{k\}$ and $E = \{k \nleftrightarrow Y \cup T_0\}$, and abbreviate

$$P_1 = \mathbb{P}(E), \quad P_2 = \mathbb{P}\big(E \cap \{o \leftrightarrow k\}\big), \quad I_1 = \mathbb{E}[F(C_k); E], \quad I_2 = \mathbb{E}\big[F(C_k); E \cap \{o \leftrightarrow k\}\big].$$

The identity (14) with $u = o$ reads $\Lambda_o(T) = \Lambda_o(T_0) + I_2 - m_k P_2$. Apply (5) with $s = k$, with $X = Y \cup T_0$ and with the two increasing functions $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$ and the indicator of the event that o lies in the cluster of k; the result is $P_2 I_1 \leq P_1 I_2$, which rearranges to

$$P_1\big(m_k P_2 - I_2\big) \ \leq \ P_2\big(m_k P_1 - I_1\big) = P_2\, \gamma_k.$$

By (16) and $P_2 \geq 0$ the right side is at most $P_2 \Lambda_k(T_0)$, and by Proposition 4.2 with $v = k$ that is at most $P_1 \Lambda_o(T_0)$. Hence

$$P_1\big(m_k P_2 - I_2\big) \ \leq \ P_1 \Lambda_o(T_0), \qquad \text{that is} \qquad P_1 \Lambda_o(T) \ \geq \ 0.$$

If $P_1 > 0$ this is the assertion. If $P_1 = 0$ then $P_2 = 0$ and $I_2 = 0$, so $\Lambda_o(T) = \Lambda_o(T_0)$, which is nonnegative by the induction hypothesis. □

The next section turns Theorem 4.1 into a statement in which the vertex realising the minimum is fixed before the event $\{o \leftrightarrow A\}$ is imposed.

5 A prescribed vertex works in Conjecture 4

The obstacle to reading Conjecture 2.4 off Theorem 4.1 is that the numbers m_x of (10) are conditional means, whereas Conjecture 2.4 compares unconditional ones. The following function removes the difference. For a vertex a and an increasing F, put

$$\Phi_a(K) = F(K) - \mathbb{E}\big[F(C_a) \text{ in a fresh configuration with every pair meeting } K \text{ deleted}\big]. \tag{18}$$

Lemma 5.1. *Let a and o be vertices and let F be increasing. Then Φ_a is increasing, and:*

(i) *for every family $\mathcal{S}$ of sets of pairs,*

$$\mathbb{E}\big[F(C_o) - F(C_a);\, o \nleftrightarrow a,\ \mathcal{C}_o \in \mathcal{S}\big] = \mathbb{E}\big[\Phi_a(C_o);\, o \nleftrightarrow a,\ \mathcal{C}_o \in \mathcal{S}\big];$$

(ii) $\mathbb{E}\big[\Phi_a(C_o);\, o \not\leftrightarrow a\big] = \mathbb{E}F(C_o) - \mathbb{E}F(C_a)$.

Proof. Enlarging K increases $F(K)$ and deletes more pairs, which decreases the subtracted mean, so Φ_a is increasing.

For (i), condition on $\mathcal{C}_o$. On $\{o \not\leftrightarrow a\}$ the vertex a lies outside C_o, every pair with exactly one endpoint in C_o is closed, and the pairs with both endpoints outside C_o retain their original independent weights. Hence the conditional mean of $F(C_a)$ given $\mathcal{C}_o$ is the mean of $F(C_a)$ in a fresh configuration with every pair meeting C_o deleted, which is $F(C_o) - \Phi_a(C_o)$. Both events in the statement are determined by $\mathcal{C}_o$, so summing over the values of $\mathcal{C}_o$ in $\mathcal{S}$ gives (i).

For (ii), take $\mathcal{S}$ to consist of all sets of pairs in (i). On $\{o \leftrightarrow a\}$ the two clusters coincide, so $F(C_o) - F(C_a)$ vanishes there and the left side of (i) equals $\mathbb{E}[F(C_o) - F(C_a)]$. □

Theorem 5.2. *Let A be a nonempty finite set of vertices, let $a \in A$, let o be a vertex and let F be increasing with $\mathbb{E}F(C_a) \le \mathbb{E}F(C_x)$ for every $x \in A$. Then*

$$\mathbb{E}\big[F(C_a);\, o \leftrightarrow A\big] \;\le\; \mathbb{E}\big[F(C_o);\, o \leftrightarrow A\big].$$

Proof. Let T be the set of $x \in A \setminus \{a\}$ with $\mathbb{P}(x \not\leftrightarrow a) > 0$. Apply Theorem 4.1 with $Y = \{a\}$, with the set T, with the increasing function Φ_a of Lemma 5.1 and with the vertex o. By Lemma 5.1(ii) applied at x in place of o, the number (10) attached to $x \in T$ is

$$\frac{\mathbb{E}\big[\Phi_a(C_x);\, x \not\leftrightarrow a\big]}{\mathbb{P}(x \not\leftrightarrow a)} = \frac{\mathbb{E}F(C_x) - \mathbb{E}F(C_a)}{\mathbb{P}(x \not\leftrightarrow a)} \;\ge\; 0,$$

so the sum subtracted in (12) is nonnegative and Theorem 4.1 gives

$$\mathbb{E}\big[\Phi_a(C_o);\, o \not\leftrightarrow a,\ o \leftrightarrow T\big] \;\ge\; 0.$$

The event $\{o \leftrightarrow T\}$ is determined by $\mathcal{C}_o$, so Lemma 5.1(i) converts this into

$$\mathbb{E}\big[F(C_o) - F(C_a);\, o \not\leftrightarrow a,\ o \leftrightarrow T\big] \;\ge\; 0. \tag{19}$$

Split $\{o \leftrightarrow A\}$ into $\{o \leftrightarrow a\}$, on which the integrand $F(C_o) - F(C_a)$ vanishes, and $\{o \not\leftrightarrow a\} \cap \{o \leftrightarrow A \setminus \{a\}\}$. The latter event differs from $\{o \not\leftrightarrow a\} \cap \{o \leftrightarrow T\}$ only by configurations in which o misses a and reaches some $x \in A \setminus \{a\}$ outside T; such a configuration lies in $\{x \not\leftrightarrow a\}$, which has probability zero for such an x, and a finite union of null events is null. Hence the left side of (19) equals $\mathbb{E}[F(C_o) - F(C_a); o \leftrightarrow A]$. □

Conjecture 2.4 follows at once by choosing a to realise the minimum of $\mathbb{E}F(C_x)$ over $x \in A$, provided every monotone cluster property is of the form $\omega \mapsto F(C_v(\omega))$ for an increasing F. That is the content of the next lemma, whose proof also shows that the first requirement in the definition of a monotone cluster property, that $f(v, \omega)$ depend only on C_v, is implied by the second half of that requirement, that $f(v, \omega)$ be increasing in C_v.

Lemma 5.3. *Let f be a function of a vertex and a configuration such that $f(v, \omega) \le f(v, \xi)$ whenever $C_v(\omega) \subseteq C_v(\xi)$ and such that $f(v, \omega) = f(w, \omega)$ whenever the pair $\{v, w\}$ is open in ω. Then there is an increasing function F of vertex sets with $f(v, \omega) = F(C_v(\omega))$ for every vertex v and every configuration ω.*

Proof. For a nonempty vertex set S let ω_S be the configuration in which the open pairs are exactly those with both endpoints in S. Every vertex of S has open cluster S in ω_S, and for distinct $v, w \in S$ the pair $\{v, w\}$ is open in ω_S, so $f(v, \omega_S)$ does not depend on the choice of $v \in S$; call the common value $F(S)$. If $S \subseteq S_1$ are nonempty and $v \in S$ then $C_v(\omega_S) = S \subseteq S_1 = C_v(\omega_{S_1})$, so $F(S) \le F(S_1)$. Extend F to the empty set by any value at most $F(S)$ for all nonempty S. Finally, for a vertex v and a configuration ω put $S = C_v(\omega)$; then $C_v(\omega_S) = S = C_v(\omega)$, and the hypothesis applied in both directions gives $f(v, \omega) = f(v, \omega_S) = F(S)$. □

Corollary 5.4. *Conjecture 2.4 holds. In particular it holds for $f(v, \omega) = |C_v|$ and for $f(v, \omega) = \mathbf{1}\{|C_v| \ge k\}$ for every k.*

Proof. By Lemma 5.3 write $f(v, \cdot) = F(C_v)$ with F increasing, choose $a \in A$ realising the minimum of $\mathbb{E}F(C_x)$ over $x \in A$, and apply Theorem 5.2. The two displayed examples are increasing functions of the vertex set. □

The remaining conjectures of the paper follow by specialising F to an indicator.

Corollary 5.5. *The inequality* (1) *holds, and so do Conjectures 2.1, 2.2 and 2.3; Conjecture 2.3 holds with $\delta = \varepsilon/2$. Question 2.7 has an affirmative answer.*

Proof. Take $F = \mathbf{1}\{b \in \cdot\}$, so that $\mathbb{E}[F(C_x); o \leftrightarrow A] = \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$ and $\mathbb{E}F(C_x) = \mathbb{P}(x \leftrightarrow b)$. Corollary 5.4 is then (1), and dropping the event $\{o \leftrightarrow A\}$ on the left gives Conjecture 2.2. If a minimises $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$ then Theorem 5.2 gives $\mathbb{P}(a \leftrightarrow b, o \leftrightarrow A) \le \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$, which is (4), so Question 2.7 has an affirmative answer.

For Conjecture 2.1, the events $\{x \leftrightarrow b\}$ and $\{o \leftrightarrow A\}$ are both increasing, so the Harris inequality gives $\mathbb{P}(o \leftrightarrow A)\mathbb{P}(x \leftrightarrow b) \le \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$ for every $x \in A$. Taking the minimum over $x \in A$ and using (1),

$$\mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b) \;\le\; \min_{x \in A} \mathbb{P}(x \leftrightarrow b,\ o \leftrightarrow A) \;\le\; \mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \;\le\; \mathbb{P}(o \leftrightarrow b).$$

For Conjecture 2.3 take $\delta = \varepsilon/2$. If $\varepsilon \ge 2$ then $1 - \varepsilon \le -1 < \mathbb{P}(o \leftrightarrow b)$ and there is nothing to prove. Otherwise $1 - \varepsilon/2$ is positive, and Conjecture 2.1 gives $\mathbb{P}(o \leftrightarrow b) > (1 - \varepsilon/2)^2 = 1 - \varepsilon + \varepsilon^2/4 \ge 1 - \varepsilon$. □

6 One coefficient vector for every target

Question 2.6 asks for one probability vector on A that works simultaneously for every vertex b. Corollary 5.4 produces, for each b, a vertex of A that works for that b; the passage from one to the other is finite-dimensional convexity. The first step is to apply Corollary 5.4 not to a single indicator but to a nonnegative weighted count of the vertices of the cluster.

Lemma 6.1. *Let A be a nonempty finite set of vertices, let o be a vertex and let $y_b \ge 0$ for $b \in V$. Then there is a vertex $a \in A$ with*

$$\sum_{b \in V} y_b \, \mathbb{P}(o \leftrightarrow A,\ a \leftrightarrow b) \;\le\; \sum_{b \in V} y_b \, \mathbb{P}(o \leftrightarrow b).$$

Proof. Put $F(K) = \sum_{b \in V} y_b \mathbf{1}\{b \in K\}$, an increasing function of vertex sets because each y_b is nonnegative. For every vertex x,

$$\mathbb{E}\big[F(C_x);\ o \leftrightarrow A\big] = \sum_{b \in V} y_b \, \mathbb{P}(x \leftrightarrow b,\ o \leftrightarrow A).$$

Let $a \in A$ realise the minimum of $\mathbb{E}[F(C_x); o \leftrightarrow A]$ over $x \in A$. By Corollary 5.4,

$$\sum_{b\in V} y_b\, \mathbb{P}(a \leftrightarrow b,\ o \leftrightarrow A) \ \le\ \sum_{b\in V} y_b\, \mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \ \le\ \sum_{b\in V} y_b\, \mathbb{P}(o \leftrightarrow b),$$

the last step because $y_b \ge 0$ and $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \le \mathbb{P}(o \leftrightarrow b)$. □

Theorem 6.2. *Question 2.6 has an affirmative answer: for every finite weighted graph, every nonempty finite $A \subseteq V$ and every vertex o there are numbers $c_x \ge 0$ for $x \in A$ with $\sum_{x\in A} c_x = 1$ such that*

$$\sum_{x\in A} c_x\, \mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b) \ \le\ \mathbb{P}(o \leftrightarrow b) \qquad \textit{for every vertex } b.$$

Proof. Regard the numbers $M(x,b) = \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$ for $x \in A$ as vectors indexed by $b \in V$, and let v be the vector with entries $\mathbb{P}(o \leftrightarrow b)$. Let K be the convex hull of the finitely many vectors $M(x,\cdot)$ for $x \in A$, a compact convex subset of the space of real vectors indexed by V, and let C be the set of differences of a point of K and a vector with nonnegative entries. The set C is convex, and it is closed because it is the sum of a compact set and a closed set.

Suppose $v \notin C$. Strict separation of the point v from the closed convex set C gives a vector y indexed by V and a real number α with $\langle y, v\rangle < \alpha \le \langle y, u\rangle$ for every $u \in C$. Fix $b \in V$ and $t \ge 0$. If $u \in C$ then $u - t\,\mathbf{e}_b \in C$, where $\mathbf{e}_b$ is the vector with entry one at b and zero elsewhere, so $\langle y, u\rangle - t\,y_b \ge \alpha$ for every $t \ge 0$, which forces $y_b \ge 0$. Applying the separating inequality at the points $M(x,\cdot)$ of $K \subseteq C$ gives $\sum_b y_b M(x,b) \ge \alpha > \sum_b y_b \mathbb{P}(o \leftrightarrow b)$ for every $x \in A$, contradicting Lemma 6.1. Hence $v \in C$: there are $c_x \ge 0$ with $\sum_{x\in A} c_x = 1$ and a vector u with nonnegative entries such that $v = \sum_{x\in A} c_x M(x,\cdot) - u$. Reading this coordinate by coordinate gives the assertion. □

7 Two vertices in place of one, and Conjecture 6

Conjecture 2.5 concerns a graph in which one pair $e = \{v,w\}$ has been forced open, while the vertex a realising the minimum is prescribed in the original graph. The following theorem is the form in which it will be proved. For a finite set R of vertices write

$$C_R = \bigcup_{s\in R} C_s,$$

so that $\{R \leftrightarrow B\}$ is the event $\{C_R \cap B \ne \varnothing\}$ and $\{R \not\leftrightarrow B\}$ its complement. Nothing is assumed about the vertices of R; the set C_R need not be connected.

Theorem 7.1. *Let A be a nonempty finite set of vertices, let $a \in A$, let v and w be vertices and let F be increasing with $\mathbb{E}F(C_a) \le \mathbb{E}F(C_x)$ for every $x \in A$. Put $R = \{v,w\}$. Then*

$$\mathbb{E}\big[F(C_R) - F(C_a);\ R \not\leftrightarrow a,\ R \leftrightarrow A \setminus \{a\}\big] \ \ge\ 0. \tag{20}$$

Section 7.1 deduces Conjecture 2.5 from Theorem 7.1, and Sections 7.2 to 7.7 prove Theorem 7.1.

7.1 Forcing the pair open

Lemma 7.2. *Let v and w be vertices, put $R = \{v,w\}$ and $e = \{v,w\}$, and let ω be a configuration in which e is closed. Then C_R takes the same value in ω and in the configuration obtained from ω by opening e; the same holds for the events $\{R \not\leftrightarrow B\}$ and $\{R \leftrightarrow B\}$ for every set B of vertices, and, on $\{R \not\leftrightarrow a\}$, for the cluster C_a.*

Proof. Opening e can only enlarge clusters, so C_R can only grow. Conversely, a path in the enlarged configuration from v or w to a vertex u that uses e splits at its uses of e into segments of ω, each of which starts at v or at w; the last segment exhibits u in $C_v \cup C_w$ computed in ω. Hence C_R is unchanged, and so are the events determined by C_R. If moreover $R \not\leftrightarrow a$ in ω then no path of ω from a meets $\{v, w\}$, so no path from a in the enlarged configuration can use e, and C_a is unchanged as well. $\square$

Lemma 7.3. *Let g be a real function of configurations that takes the same value on a configuration in which e is closed and on the configuration obtained from it by opening e. Then the mean of g under $\mathbb{P}^e$ equals its mean under $\mathbb{P}$.*

Proof. Partition the configurations into the pairs consisting of a configuration η in which e is closed and the configuration obtained from η by opening e. Under $\mathbb{P}$ the two members of such a pair have probabilities $(1 - p_e)\rho(\eta)$ and $p_e\rho(\eta)$, where $\rho(\eta)$ is the product over the pairs other than e of the corresponding factor; under $\mathbb{P}^e$ they have probabilities 0 and $\rho(\eta)$. Since g takes the same value at both members, each pair contributes $\rho(\eta)g(\eta)$ to both means. $\square$

Theorem 7.4. *Let A be a nonempty finite set of vertices, let b be a vertex, let $a \in A$ minimise $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$, and let $e = \{v, w\}$ be a pair. Then*

$$\mathbb{P}^e(a \leftrightarrow b,\ v \leftrightarrow A) \ \leq\ \mathbb{P}^e(v \leftrightarrow b,\ v \leftrightarrow A), \tag{21}$$

and Conjecture 2.5 holds. Its hypothesis (2) *is a consequence of Conjecture 2.1 and therefore carries no information.*

Proof. Apply Theorem 7.1 with $F = \mathbf{1}\{b \in \cdot\}$, for which $\mathbb{E}F(C_x) = \mathbb{P}(x \leftrightarrow b)$, so that the hypothesis on a is exactly the minimality assumed here. By Lemmas 7.2 and 7.3 the left side of (20) has the same value under $\mathbb{P}$ and under $\mathbb{P}^e$, since the integrand and the two events defining the range of integration are unchanged by opening e on the range of integration. Hence

$$\mathbb{E}^e\big[F(C_R) - F(C_a);\ R \not\leftrightarrow a,\ R \leftrightarrow A \setminus \{a\}\big] \ \geq\ 0, \qquad R = \{v, w\}.$$

Under $\mathbb{P}^e$ the pair e is open with probability one, so almost surely $C_v = C_w = C_R$, the event $\{R \not\leftrightarrow a\}$ is $\{v \not\leftrightarrow a\}$ and the event $\{R \leftrightarrow A \setminus \{a\}\}$ is $\{v \leftrightarrow A \setminus \{a\}\}$. Splitting $\{v \leftrightarrow A\}$ into $\{v \leftrightarrow a\}$, on which $C_v = C_a$ and the integrand vanishes, and $\{v \not\leftrightarrow a\} \cap \{v \leftrightarrow A \setminus \{a\}\}$, the last display becomes

$$\mathbb{E}^e\big[\mathbf{1}\{b \in C_v\} - \mathbf{1}\{b \in C_a\};\ v \leftrightarrow A\big] \ \geq\ 0,$$

which is (21). The events $\{a \leftrightarrow b\}$ and $\{v \leftrightarrow A\}$ are increasing, so the Harris inequality and (21) give

$$\mathbb{P}^e(v \leftrightarrow A)\mathbb{P}^e(a \leftrightarrow b) \ \leq\ \mathbb{P}^e(a \leftrightarrow b,\ v \leftrightarrow A) \ \leq\ \mathbb{P}^e(v \leftrightarrow b,\ v \leftrightarrow A) \ \leq\ \mathbb{P}^e(v \leftrightarrow b),$$

which is (3). Finally, Conjecture 2.1 applied with x in place of o gives $\mathbb{P}(x \leftrightarrow b) \geq \mathbb{P}(x \leftrightarrow A) \min_{y \in A} \mathbb{P}(y \leftrightarrow b) = \mathbb{P}(x \leftrightarrow A)\mathbb{P}(a \leftrightarrow b)$, which is (2). $\square$

7.2 Events conditioned on a set missing another set

Throughout Sections 7.2 to 7.5 fix a vertex x, a set Y of vertices with $x \notin Y$, and an increasing function g of sets of pairs, and write $D = \{x \not\leftrightarrow Y\}$. For finite sets R, X and B of vertices put

$$H(R; X, B) = \{R \not\leftrightarrow B\} \cap \{R \leftrightarrow X\}, \tag{22}$$

the event that C_R misses B and meets X. Two features distinguish (22) from the events used in Section 4. First, $\{R \leftrightarrow x\}$ does not imply $\{R \not\leftrightarrow Y\}$ even when $\{x \not\leftrightarrow Y\}$ holds, because a component of C_R other than the one containing x may meet Y; the two conditions must therefore be carried together. Second, $H(\{u\}; \{x\}, Y) = D \cap \{x \leftrightarrow u\}$ for every vertex u, since $u \leftrightarrow x$ makes the clusters of u and x equal. Set

$$\Gamma(R) = \mathbb{P}(D)\,\mathbb{E}\big[g(\mathcal{C}_x);\, H(R; \{x\}, Y)\big] - \mathbb{E}\big[g(\mathcal{C}_x);\, D\big]\,\mathbb{P}\big(H(R; \{x\}, Y)\big), \tag{23}$$

which at $R = \{u\}$ is $\mathbb{P}(D)^2$ times the conditioned covariance appearing in Theorem 3.2. The event $H(R; \{x\}, Y)$ is contained in D: on it x lies in C_R and C_R misses Y, so $C_x \subseteq C_R$ misses Y.

The functional (7) is now applied to functions of a set argument rather than of a vertex. Fix distinct vertices $z_1, \ldots, z_\ell$ and a further vertex v, all outside $\{x\} \cup Y$ and distinct from each other, put $B_1 = \{x\} \cup Y$ and $B_{j+1} = B_j \cup \{z_j\}$, and put

$$c_j(R) = \frac{\mathbb{P}\big(H(R; \{z_j\}, B_j)\big)}{\mathbb{P}(z_j \not\leftrightarrow B_j)}, \qquad \tau(R) = \frac{\mathbb{P}\big(H(R; \{v\}, B_{\ell+1})\big)}{\mathbb{P}(v \not\leftrightarrow B_{\ell+1})}. \tag{24}$$

For a real function q of finite vertex sets put $q^0 = q$, then $q^j(R) = q^{j-1}(R) - c_j(R)q^{j-1}(\{z_j\})$ for $1 \le j \le \ell$, and finally, for a prescribed set R_0,

$$\mathrm{M}[q] = q^\ell(R_0) - \tau(R_0)\, q^\ell(\{v\}). \tag{25}$$

At $R_0 = \{u\}$ the numbers (24) are the constants (6) and (25) is (7), so $\mathrm{M}[\Gamma] \ge 0$ is Theorem 3.2. The goal of Sections 7.2 to 7.5 is the same conclusion for $R_0 = \{s_1, s_2\}$ with $s_1 \ne s_2$.

Lemma 7.5. *There are real numbers λ_u for $u \in V$, depending only on $z_1, \ldots, z_\ell$, v and R_0, such that $\mathrm{M}[q] = q(R_0) + \sum_{u \in V} \lambda_u\, q(\{u\})$ for every real function q of finite vertex sets.*

Proof. Induct on ℓ. For $\ell = 0$ the assertion holds with $\lambda_v = -\tau(R_0)$ and all other λ_u zero. Passing from $\ell - 1$ to ℓ replaces q by $R \mapsto q(R) - c_1(R)q(\{z_1\})$; the coefficient of $q(R_0)$ is unchanged, and the extra term is a multiple of $q(\{z_1\})$. Collecting equal vertices gives the assertion. □

Only the value at R_0 enters with a coefficient that is not attached to a singleton, and that coefficient is one. This is what makes the one-sided estimates below usable.

7.3 Exploring the cluster of a set

Let z be a vertex, let B be a set of vertices containing $\{x\} \cup Y$ but not z, and let Θ and Ψ be as in (8) and the paragraph containing it, built from x, Y and g. Abbreviate $H_R = H(R; \{z\}, B)$ and $E = \{z \not\leftrightarrow B\}$, and assume $\mathbb{P}(E) > 0$.

Lemma 7.6. *For every finite set R of vertices, $\mathbb{E}[\Theta; H_R] = -\mathbb{E}[\Psi(C_R); H_R]$.*

Proof. Sum over the possible values of the complete set of open pairs met by paths from R. On H_R the vertex set C_R misses B, hence misses $\{x\} \cup Y$, and every pair with exactly one endpoint in C_R is closed. Given that complete data, the pairs with both endpoints outside C_R retain their original independent weights, and deleting every pair meeting C_R changes neither $\mathcal{C}_x$ nor $\mathcal{C}_Y$ nor the quantity $\bar{f}$ of (8). The conditional mean of Θ is therefore the mean of Θ in a configuration with every pair meeting C_R deleted, which is $-\Psi(C_R)$ by definition. The event H_R is determined by the same data, so summing over its values gives the assertion. □

Lemma 7.7. *Put* $\Delta(R) = \mathbb{E}[\Psi(C_R) - \Psi(C_z); H_R]$*. Then* $\Delta(R) \geq 0$ *for every finite* R*, and* $\Delta(\{u\}) = 0$ *for every vertex* u*.*

Proof. On H_R some vertex of R is joined to z, so $C_z \subseteq C_R$ and $\Psi(C_z) \leq \Psi(C_R)$ because Ψ is increasing. If $R = \{u\}$ then on $H_{\{u\}}$ the clusters of u and z coincide, so the integrand vanishes identically. □

The next inequality is the step that transfers the estimate from a single vertex to a set. Let $\Gamma_{z,B}$ denote the expression (23) formed with the vertex z, the set B and the increasing function $\mathcal{C} \mapsto \Psi(\{z\} \cup \operatorname{span} \mathcal{C})$ in place of x, Y and g.

Lemma 7.8. *For every finite set* R *of vertices,*

$$\mathbb{E}[\Theta; H_R] - c(R)\,\mathbb{E}[\Theta; E] = -\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)} - \Delta(R), \qquad c(R) = \frac{\mathbb{P}(H_R)}{\mathbb{P}(E)}, \tag{26}$$

and consequently

$$\mathbb{E}[\Theta; H_R] - c(R)\,\mathbb{E}[\Theta; E] \;\leq\; -\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)}, \tag{27}$$

with equality in (27) *whenever* R *is a single vertex.*

Proof. Expanding (23) for the vertex z, the set B and the stated function, and using that $\Psi(\{z\} \cup \operatorname{span} \mathcal{C}_z) = \Psi(C_z)$,

$$\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)} = \mathbb{E}[\Psi(C_z); H_R] - c(R)\,\mathbb{E}[\Psi(C_z); E].$$

By Lemma 7.6 and the definition of Δ, $\mathbb{E}[\Theta; H_R] = -\mathbb{E}[\Psi(C_z); H_R] - \Delta(R)$, and by (9), $\mathbb{E}[\Theta; E] = -\mathbb{E}[\Psi(C_z); E]$. Substituting gives (26). The inequality (27) follows from $\Delta \geq 0$, with equality at a single vertex by Lemma 7.7. □

7.4 Transfer between two vertex sets in a restricted graph

For a set $U \subseteq V$ write $\mathbb{P}_U$ for percolation in which every pair with an endpoint outside U is closed and every pair with both endpoints in U keeps its weight. Every quantity below formed inside U is set to zero if one of the distinguished vertices appearing in it lies outside U, and every set appearing in it is intersected with U. Fix a set X of vertices and a vertex $v \notin X$, and let R be a finite set of vertices disjoint from $X \cup \{v\}$. For a set K of vertices put

$$\alpha_{R,B}(K) = \mathbf{1}\{K \cap R \neq \varnothing\}\; \mathbb{P}_{U \setminus K}\big(R \setminus K \nleftrightarrow B\big). \tag{28}$$

Lemma 7.9. *The function* $K \mapsto \alpha_{R,B}(K)$ *is increasing, the function* $B \mapsto \alpha_{R,B}(K)$ *is decreasing, and if* $\{z\} \cup R \cup B \subseteq U$ *then*

$$\mathbb{E}_U\big[\alpha_{R,B}(C_v);\, v \nleftrightarrow B\big] = \mathbb{P}_U\big(R \nleftrightarrow B,\; R \leftrightarrow v\big). \tag{29}$$

Proof. Enlarging K can only turn the first factor from zero to one; it also removes vertices from U and from R, so every path from $R \setminus K$ to B in $U \setminus K$ survives the enlargement, and the avoidance probability increases. Enlarging B decreases that probability. For (29), sum over the possible values of the complete set of open pairs met by paths from v. On $\{v \nleftrightarrow B\}$ the vertex set $K = C_v$ misses B, the event $\{R \leftrightarrow v\}$ is $\{K \cap R \neq \varnothing\}$, every pair with exactly one endpoint in K is closed, and the pairs with both endpoints outside K are fresh. On that data the event $\{R \nleftrightarrow B\}$ is the event that $R \setminus K$ misses B in $U \setminus K$, since the vertices of R inside K have cluster C_v, which misses B. □

Lemma 7.10. *Let N and N_1 be disjoint sets of vertices, and let all the sets and vertices named lie in U. Then*

$$\mathbb{P}_U\big(R \nleftrightarrow X \cup N,\ R \leftrightarrow v\big)\,\mathbb{P}_U\big(v \nleftrightarrow X \cup N_1\big) \ \le\ \mathbb{P}_U\big(v \nleftrightarrow X \cup N \cup N_1\big)\,\mathbb{P}_U\big(R \nleftrightarrow X,\ R \leftrightarrow v\big). \quad (30)$$

Proof. By (29) the left side is $\mathbb{E}_U[\alpha_{R,X\cup N}(C_v); v \nleftrightarrow X \cup N]\,\mathbb{P}_U(v \nleftrightarrow X \cup N_1)$. Apply Theorem 3.1 inside U with the vertex v, the two sets $X \cup N$ and $X \cup N_1$, whose intersection is X and whose union is $X \cup N \cup N_1$ because N and N_1 are disjoint, and the two increasing functions $\alpha_{R,X\cup N}$ and the constant one. The result is at most $\mathbb{E}_U[\alpha_{R,X\cup N}(C_v); v \nleftrightarrow X]\,\mathbb{P}_U(v \nleftrightarrow X \cup N \cup N_1)$. By Lemma 7.9 the first factor is at most $\mathbb{E}_U[\alpha_{R,X}(C_v); v \nleftrightarrow X]$, which is $\mathbb{P}_U(R \nleftrightarrow X, R \leftrightarrow v)$ by (29). If v lies in an avoided set or R meets one, the left side is zero. □

Lemma 7.11. *For a nonnegative real function h of subsets of U that vanishes on sets not containing v, and for sets N and N_1 of vertices contained in U, put*

$$Y_h(N) = \mathbb{E}_U\big[\mathbf{1}\{x \nleftrightarrow N\}\,h(U \setminus C_N)\big], \qquad X_h(N) = \mathbb{E}_U\big[\mathbf{1}\{x \nleftrightarrow N\}\mathbf{1}\{R \nleftrightarrow N\}\,\theta(U \setminus C_N)\,h(U \setminus C_N)\big],$$

where $\theta(W) = \mathbb{P}_W(R \nleftrightarrow X, R \leftrightarrow v)/\mathbb{P}_W(v \nleftrightarrow X)$. Assume $x \in X$, $v \notin X$, that R is disjoint from $X \cup \{v\}$, that all the named sets and vertices lie in U, that in every $W \subseteq U$ and for every $Q \subseteq W$ one has $Y_h(Q)\mathbb{P}_W(v \nleftrightarrow X) \le \mathbb{P}_W(v \nleftrightarrow X \cup Q)h(W)$ with Y_h formed inside W, and that $Q \mapsto Y_h(Q)$ is decreasing in every such W. Then

$$\mathbb{P}_U\big(R \nleftrightarrow X \cup N,\ R \leftrightarrow v\big)\,Y_h(N_1) \ \le\ \mathbb{P}_U\big(v \nleftrightarrow X \cup N \cup N_1\big)\,X_h(N \cap N_1). \quad (31)$$

Proof. Induct on the number of elements of U. If N and N_1 are disjoint, multiply the assumed inequality for N_1 by $\mathbb{P}_U(R \nleftrightarrow X \cup N, R \leftrightarrow v)$ and apply Lemma 7.10, using $\mathbb{P}_U(R \nleftrightarrow X, R \leftrightarrow v) = \theta(U)\mathbb{P}_U(v \nleftrightarrow X)$ and $X_h(\varnothing) = \theta(U)h(U)$. If $\mathbb{P}_U(v \nleftrightarrow X) = 0$ then $\mathbb{P}_U(R \nleftrightarrow X \cup N, R \leftrightarrow v) = 0$ by event inclusion and both sides vanish.

Otherwise put $Z = N \cap N_1$, which is nonempty. If $v \in Z$, or $x \in Z$, or R meets Z, the left side is zero. In the remaining case remove Z and expose the set $\sigma \subseteq U \setminus Z$ of vertices joined to Z by an open pair. The indicators of the events $\{u \in \sigma\}$ are determined by disjoint blocks of pairs, so σ has a product law ν on subsets of $U \setminus Z$. With $N_2 = N \setminus Z$ and $N_3 = N_1 \setminus Z$, the four quantities in (31) are the ν-averages of

$$e(\sigma) = \mathbb{P}_{U\setminus Z}\big(R \nleftrightarrow X \cup N_2 \cup \sigma,\ R \leftrightarrow v\big), \quad y(\sigma) = Y_h(N_3 \cup \sigma), \quad m(\sigma) = \mathbb{P}_{U\setminus Z}\big(v \nleftrightarrow X \cup N_2 \cup N_3 \cup \sigma\big),$$

and $\chi(\sigma) = X_h(\sigma)$, all formed inside $U \setminus Z$: the event that R misses N in U is the event that R misses $N_2 \cup \sigma$ in $U \setminus Z$, and likewise for the other three, so no factor is taken outside an expectation. For subsets S and S_1 of $U \setminus Z$ put $P = S \cup (N_2 \setminus S_1)$ and $P_1 = S_1 \cup (N_3 \setminus S)$; then $P \cup P_1 = N_2 \cup N_3 \cup S \cup S_1$ and $P \cap P_1 = S \cap S_1$, so the induction hypothesis inside $U \setminus Z$, together with the assumed monotonicity of Y_h, gives $e(S)y(S_1) \le m(S \cup S_1)\chi(S \cap S_1)$. Multiplying by the product identity $\nu(S)\nu(S_1) = \nu(S \cap S_1)\nu(S \cup S_1)$ and applying the Ahlswede and Daykin four functions theorem to the four functions $\sigma \mapsto \nu(\sigma)e(\sigma)$, $\sigma \mapsto \nu(\sigma)y(\sigma)$, $\sigma \mapsto \nu(\sigma)m(\sigma)$ and $\sigma \mapsto \nu(\sigma)\chi(\sigma)$ yields (31). □

Lemma 7.12. *Let $x \in X$, let $v \notin X$, let R be disjoint from $X \cup \{v\}$ and let g be increasing. Put $\eta_R = \mathrm{Cov}_U\big(g(\mathcal{C}_x), \mathbf{1}\{R \leftrightarrow X\}\big)$ and $\eta_v = \mathrm{Cov}_U\big(g(\mathcal{C}_x), \mathbf{1}\{v \leftrightarrow X\}\big)$. Then $\eta_v \ge 0$ and*

$$\theta(U)\,\eta_v \ \le\ \eta_R. \quad (32)$$

Proof. All expectations are taken under $\mathbb{P}_U$. The events $\{v \leftrightarrow X\}$ and $\{g(\mathcal{C}_x) \geq t\}$ are increasing, so $\eta_v \geq 0$ by the Harris inequality. Put $J = \{R \not\leftrightarrow X\} \cap \{R \leftrightarrow v\}$ and $D_1 = \{v \not\leftrightarrow X\}$. If $v \leftrightarrow X$ and $R \leftrightarrow v$ then $R \leftrightarrow X$, so $J \subseteq D_1$, and $\mathbf{1}\{R \leftrightarrow X\} + \mathbf{1}_J = \mathbf{1}\{R \leftrightarrow X \text{ or } R \leftrightarrow v\}$, an increasing event. The Harris inequality applied to that event gives $\eta_R + \mathrm{Cov}_U(g(\mathcal{C}_x), \mathbf{1}_J) \geq 0$, that is

$$\eta_R \;\geq\; \mathbb{E}_U[g(\mathcal{C}_x)]\,\mathbb{P}_U(J) - \mathbb{E}_U[g(\mathcal{C}_x); J].$$

Conditionally on D_1 the set of open pairs met by paths from X and the set of open pairs met by paths from v are disjoint; the function $g(\mathcal{C}_x)$ is increasing in the first and constant in the second, while $\mathbf{1}_J$ is decreasing in the first and increasing in the second. The second part of Theorem 3.1, applied to $g(\mathcal{C}_x)$ and $-\mathbf{1}_J$, gives $\mathbb{P}_U(D_1)\mathbb{E}_U[g(\mathcal{C}_x); J] \leq \mathbb{P}_U(J)\mathbb{E}_U[g(\mathcal{C}_x); D_1]$. Since $\eta_v = \mathbb{E}_U[g(\mathcal{C}_x)]\mathbb{P}_U(D_1) - \mathbb{E}_U[g(\mathcal{C}_x); D_1]$ and $\theta(U) = \mathbb{P}_U(J)/\mathbb{P}_U(D_1)$,

$$\theta(U)\eta_v = \mathbb{E}_U[g(\mathcal{C}_x)]\mathbb{P}_U(J) - \frac{\mathbb{P}_U(J)\,\mathbb{E}_U[g(\mathcal{C}_x); D_1]}{\mathbb{P}_U(D_1)} \;\leq\; \mathbb{E}_U[g(\mathcal{C}_x)]\mathbb{P}_U(J) - \mathbb{E}_U[g(\mathcal{C}_x); J] \;\leq\; \eta_R.$$

If $\mathbb{P}_U(D_1) = 0$ then $\eta_v = 0$ and the assertion is the Harris inequality. $\square$

Proposition 7.13. *Let every pair weight lie strictly between zero and one, let $X = \{x\} \cup \{z_1, \ldots, z_\ell\}$, let $v \notin X \cup Y$, let R_0 be disjoint from $X \cup Y \cup \{v\}$ and let g be increasing. Write $U = V \setminus C_Y$ and let η_{R_0} and η_v be as in Lemma 7.12. Then*

$$\mathcal{H} := \mathbb{E}\big[\mathbf{1}_D \mathbf{1}\{R_0 \not\leftrightarrow Y\}\,\eta_{R_0}\big] - \tau(R_0)\,\mathbb{E}\big[\mathbf{1}_D\,\eta_v\big] \;\geq\; 0. \tag{33}$$

Proof. Subtracting from g its minimum over the finitely many values of $\mathcal{C}_x$ changes neither η_{R_0} nor η_v, so assume $g \geq 0$. Take $h(W) = \mathrm{Cov}_W(g(\mathcal{C}_x), \mathbf{1}\{v \leftrightarrow X\})$ for $W \subseteq V$, extended by zero on sets not containing v; then $h \geq 0$ by the Harris inequality. The hypotheses of Lemma 7.11 for this h are exactly the stopped form of the Harris inequality for the conditioned covariance and its monotonicity in the set that is avoided. Apply Lemma 7.11 with $R = R_0$ and $N = N_1 = Y$, whose intersection and union are both Y. Its left side is $\mathbb{P}(R_0 \not\leftrightarrow X \cup Y, R_0 \leftrightarrow v)\,\mathbb{E}[\mathbf{1}_D \eta_v]$ and its right side is $\mathbb{P}(v \not\leftrightarrow X \cup Y)\,\mathbb{E}[\mathbf{1}_D \mathbf{1}\{R_0 \not\leftrightarrow Y\}\theta(U)\eta_v]$. Dividing by $\mathbb{P}(v \not\leftrightarrow X \cup Y)$, which is positive because the configuration with no open pair has positive probability, and recognising $\tau(R_0)$ as the resulting quotient by (24) with $B_{\ell+1} = X \cup Y$,

$$\tau(R_0)\,\mathbb{E}\big[\mathbf{1}_D\,\eta_v\big] \;\leq\; \mathbb{E}\big[\mathbf{1}_D \mathbf{1}\{R_0 \not\leftrightarrow Y\}\,\theta(U)\,\eta_v\big].$$

By Lemma 7.12 applied inside U, the integrand on the right is at most $\mathbf{1}_D \mathbf{1}\{R_0 \not\leftrightarrow Y\}\eta_{R_0}$. $\square$

7.5 The hierarchy for a set of two vertices

Two quantities organise the induction. Recall Θ from (8) and put

$$\mathcal{W} = \mathbb{E}\big[\Theta \cdot \mathrm{M}[\beta]\big], \tag{34}$$

where $\beta(R) = \mathbf{1}\{R \cap C_Y = \varnothing\}\mathbf{1}\{R \leftrightarrow X\}$ with $X = \{x\} \cup \{z_1, \ldots, z_\ell\}$, and where M acts on the argument R with the values of $\mathcal{C}_x$ and $\mathcal{C}_Y$ held fixed. Because $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$, the quantity $\mathcal{W}$ is the average over $\mathcal{C}_Y$ of a conditional covariance, and it is the part of $\mathrm{M}[\Gamma]$ that is produced after the cluster of Y has been exposed.

Proposition 7.14. *Let every pair weight lie strictly between zero and one, let $z_1, \dots, z_\ell$, v and R_0 be as in Proposition 7.13, and let g be increasing. Then, with M_j the functional* (25) *built from $z_{j+1}, \dots, z_\ell$, v and R_0 with base set B_{j+1},*

$$\mathcal{W} = \mathcal{H} + \sum_{j=1}^{\ell} \frac{\mathrm{M}_j\left[\Gamma_{z_j,B_j}\right]}{\mathbb{P}(z_j \not\leftrightarrow B_j)} + \sum_{j=1}^{\ell} \Delta_j(R_0), \tag{35}$$

where Γ_{z_j,B_j} and Δ_j are the quantities of Section 7.3 for the vertex z_j and the set B_j.

Proof. Write $Z = C_Y$ and, for a set S of vertices and a vertex z,

$$\beta_S(R) = \mathbf{1}\{R \cap Z = \varnothing,\ R \leftrightarrow S\}, \qquad \delta_{S,z}(R) = \mathbf{1}\{R \cap Z = \varnothing,\ R \not\leftrightarrow S,\ R \leftrightarrow z\}.$$

Two pointwise identities hold: $\beta_{S \cup \{z\}} = \beta_S + \delta_{S,z}$, and $\beta_S(\{z\}) = \mathbf{1}\{z \notin Z\} - \delta_{S,z}(\{z\})$, the latter because connection is reflexive. Hence, for the operator $q \mapsto q(\cdot) - c(\cdot)q(\{z\})$ with c as in (24),

$$\beta_S(R) - c(R)\beta_S(\{z\}) = \beta_{S\cup\{z\}}(R) - \big(\delta_{S,z}(R) - c(R)\delta_{S,z}(\{z\})\big) - c(R)\mathbf{1}\{z \notin Z\}.$$

Multiply by Θ and take expectations. The last term is $c(R)$ times a function of $\mathcal{C}_Y$, so its contribution vanishes. With $S = \{x\} \cup \{z_1, \dots, z_{j-1}\}$ and $z = z_j$, the event $\{R \cap Z = \varnothing, R \not\leftrightarrow S\}$ is $\{R \not\leftrightarrow B_j\}$, and $\delta_{S,z_j}(\{z_j\})$ is the indicator of $\{z_j \not\leftrightarrow B_j\}$; the middle bracket is therefore the left side of (26) for the vertex z_j and the set B_j. Applying (26),

$$\mathbb{E}\big[\Theta\big(\beta_S(R) - c(R)\beta_S(\{z_j\})\big)\big] = \mathbb{E}\big[\Theta\, \beta_{S\cup\{z_j\}}(R)\big] + \frac{\Gamma_{z_j,B_j}(R)}{\mathbb{P}(z_j \not\leftrightarrow B_j)} + \Delta_j(R).$$

Iterate from $j = 1$ to $j = \ell$ and then subtract $\tau(R_0)$ times the value at $\{v\}$. By Lemma 7.5 the terms produced at step j are evaluated at R_0 with coefficient one and at singletons otherwise, and Δ_j vanishes at every singleton by Lemma 7.7, so only $\Delta_j(R_0)$ survives among the correction terms. After all ℓ steps the remaining test function is β_X, and

$$\mathbb{E}\big[\Theta\big(\beta_X(R_0) - \tau(R_0)\beta_X(\{v\})\big)\big] = \mathcal{H},$$

because conditioning on $\mathcal{C}_Y$ and using $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$ turns each of the two terms into the conditional covariance of $g(\mathcal{C}_x)$ with the corresponding connection indicator inside $U = V \setminus C_Y$, which are η_{R_0} and η_v of Lemma 7.12, carrying the factors $\mathbf{1}_D\mathbf{1}\{R_0 \not\leftrightarrow Y\}$ and $\mathbf{1}_D\mathbf{1}\{v \notin C_Y\}$. This is (35). $\square$

Proposition 7.15. *Let every pair weight lie strictly between zero and one, let $s_1 \neq s_2$, let $R_0 = \{s_1, s_2\}$, and let $z_1, \dots, z_\ell$, v, x and Y satisfy the disjointness conditions of Section 7.2 with R_0 disjoint from $\{x, v\} \cup Y \cup \{z_1, \dots, z_\ell\}$. Then $\mathrm{M}[\Gamma] \geq 0$ for every increasing g.*

Proof. Induct on ℓ, the statement being asserted at each ℓ for all admissible data simultaneously, and prove at each step both that $\mathcal{W} \geq 0$ for every increasing g and that $\mathrm{M}[\Gamma] \geq 0$ for every increasing g.

For the first assertion, replace g by g minus its minimum, apply Proposition 7.14 and estimate the three groups of terms in (35). The first is nonnegative by Proposition 7.13. Each denominator $\mathbb{P}(z_j \not\leftrightarrow B_j)$ is positive because $z_j \notin B_j$ and the configuration with no open pair has positive probability. Each $\mathrm{M}_j[\Gamma_{z_j,B_j}]$ is a quantity of the same shape with a strictly shorter list of vertices, and Ψ is nonnegative and increasing, so it is nonnegative by the induction hypothesis. Each $\Delta_j(R_0)$ is nonnegative by Lemma 7.7. Hence $\mathcal{W} \geq 0$.

For the second assertion, expand Γ. By (23) and the containment $H(R;\{x\},Y) \subseteq D$, and writing the two indicators of $H(R;\{x\},Y)$ in terms of the vertex carriers $\{x\} \cup \operatorname{span} \mathcal{C}_x$ and $Y \cup \operatorname{span} \mathcal{C}_Y$,

$$\Gamma(R) = \mathbb{P}(D)\, \mathbb{E}\big[g(\mathcal{C}_x)\, \beta_*(R);\, D\big] - \mathbb{E}\big[g(\mathcal{C}_x); D\big]\, \mathbb{E}\big[\beta_*(R); D\big],$$

in which $\beta_*(R)$ is the indicator that R meets $\{x\} \cup \operatorname{span} \mathcal{C}_x$ and misses $Y \cup \operatorname{span} \mathcal{C}_Y$. Since M is linear in its argument and its coefficients are constants, $\mathrm{M}[\Gamma]$ is the same expression with $\beta_*(R)$ replaced by $\mathrm{M}[\beta_*]$, a function of the two sets $\mathcal{C}_x$ and $\mathcal{C}_Y$. Modify $\mathrm{M}[\beta_*]$ off the event D by dropping the second indicator at every singleton evaluation; on D the two clusters are disjoint and nothing changes, while the modified function is decreasing in $\mathcal{C}_Y$ because by Lemma 7.5 only the evaluation at R_0 depends on $\mathcal{C}_Y$ and it enters with coefficient one. Here the hypothesis $s_1 \neq s_2$ is used, since R_0 must not itself be a singleton.

The event D is the event that the two sets $\{x\}$ and Y are not joined, so the conditioned covariance of an increasing function of $\mathcal{C}_x$ with a function of the pair $(\mathcal{C}_x, \mathcal{C}_Y)$ splits, by one step of the Markov chain which resamples the pairs inside the cluster of $\{x\}$ and then those inside the cluster of Y, into the average of the conditional covariance given $\mathcal{C}_Y$, which is $\mathcal{W}$, the average of the conditional covariance given $\mathcal{C}_x$, and the same conditioned covariance with the first function replaced by its one-step average. The second of the three is nonnegative by the Harris inequality for two decreasing functions, because the one-step average of an increasing function of $\mathcal{C}_x$ is decreasing in $\mathcal{C}_Y$ and the modified test function is decreasing in $\mathcal{C}_Y$. The first is $\mathcal{W}$, which has just been shown to be nonnegative for every increasing g, and the one-step average of an increasing function is again increasing. Iterating the splitting, the remainder tends to zero because the one-step average contracts towards a constant, whence $\mathrm{M}[\Gamma] \geq 0$. □

7.6 Removing the vertices of the set one at a time

Fix a set Y, an increasing function F of vertex sets, a finite $T \subseteq V \setminus Y$ with $\mathbb{P}(x \nleftrightarrow Y) > 0$ for $x \in T$, and an injective r on T with $m_x \leq m_y$ whenever $r(x) < r(y)$, the numbers m_x being those of (10). For a finite set R of vertices define, in analogy with (11) and (12),

$$J_x(R) = \{R \nleftrightarrow Y\} \cap \{R \leftrightarrow x\} \cap \bigcap_{y \in T,\ r(y) < r(x)} \{R \nleftrightarrow y\},$$

$$\Lambda_R(T) = \mathbb{E}\big[F(C_R);\, R \nleftrightarrow Y,\ R \leftrightarrow T\big] - \sum_{x \in T} \mathbb{P}\big(J_x(R)\big)\, m_x.$$

For $R = \{u\}$ these are (11) and (12).

Theorem 7.16. *Let V be a finite vertex set with pair weights in $[0,1]$, let Y, F, T and r be as above, and let R be a set of at most two vertices, disjoint from $Y \cup T$. Then $\Lambda_R(T) \geq 0$.*

Proof. Assume first that every pair weight lies strictly between zero and one. If R is a single vertex the assertion is Theorem 4.1, so let $R = \{s_1, s_2\}$ with $s_1 \neq s_2$.

Three preliminary facts are needed. First, let $k \notin T$ satisfy $r(y) < r(k)$ for every $y \in T$, and put $E = \{k \nleftrightarrow Y \cup T\}$ and $H_R = H(R; \{k\}, Y \cup T)$. Adding k leaves the events $J_x(R)$ for $x \in T$ unchanged and creates the single new event H_R, so

$$\Lambda_R(T \cup \{k\}) - \Lambda_R(T) = \mathbb{E}\big[F(C_R) - m_k;\, H_R\big], \tag{36}$$

and on H_R some vertex of R is joined to k, so $C_k \subseteq C_R$ and the right side is at least $\mathbb{E}[F(C_k) - m_k; H_R]$, with equality when R is a single vertex. Second, with $\gamma_k = m_k \mathbb{P}(E) - \mathbb{E}[F(C_k); E]$ and with $\Gamma_{k,Y\cup T}$ the quantity (23) formed with the vertex k, the set $Y \cup T$ and the increasing function $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$,

$$\mathbb{P}(E)\,\mathbb{E}\big[F(C_k) - m_k; H_R\big] = \Gamma_{k,Y\cup T}(R) - \gamma_k\,\mathbb{P}(H_R), \tag{37}$$

as one sees by expanding both sides. Third, exactly as in the proof of Proposition 4.2,

$$\gamma_k \;\le\; \Lambda_{\{k\}}(T). \tag{38}$$

Let now M be the functional (25) with $R_0 = R$, with the vertices $z_1, \ldots, z_\ell$ and v outside $Y \cup T \cup R$, and with the sets (24) based at $Y \cup T$. We claim that

$$\mathrm{M}\big[\, S \mapsto \Lambda_S(T)\,\big] \;\ge\; 0 \tag{39}$$

for every such choice, by induction on the cardinality of T. For $T = \varnothing$ every value vanishes. Otherwise let $k \in T$ have the largest value of r, put $T_0 = T \setminus \{k\}$, and read (36) for T_0 and k. By Lemma 7.5 the value at R enters M with coefficient one and every other evaluation is at a singleton, where (36) is an equality; hence (36) may be inserted inside M with the correct sign. Multiplying by $\mathbb{P}(E) > 0$ and using (37),

$$\mathbb{P}(E)\,\mathrm{M}\big[\Lambda.(T)\big] \;\ge\; \mathbb{P}(E)\,\mathrm{M}\big[\Lambda.(T_0)\big] + \mathrm{M}\big[\Gamma_{k,Y\cup T_0}\big] - \gamma_k\,\mathbb{P}(E)\,\mathrm{M}\big[c\big],$$

where $c(S) = \mathbb{P}(H(S;\{k\}, Y \cup T_0))/\mathbb{P}(E)$. The middle term is nonnegative by Proposition 7.15 applied to the vertex k, the set $Y \cup T_0$ and the increasing function $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$. The last margin is nonnegative: Proposition 7.15 applied to the same vertex and set and to the increasing function $\mathcal{C} \mapsto \mathbf{1}\{\mathcal{C} \neq \varnothing\}$ gives $\mathrm{M}[\Gamma^*_{k,Y\cup T_0}] \ge 0$ for the corresponding quantity Γ^*, and a direct expansion gives $\Gamma^*(S) = \rho\,\mathbb{P}(H(S;\{k\}, Y \cup T_0))$ with ρ the probability that k misses $Y \cup T_0$ and lies in no open pair, a positive number, since on $H(S;\{k\}, Y \cup T_0)$ the vertex k is joined to a different vertex and therefore lies in an open pair. Dividing by $\rho\mathbb{P}(E)$ gives $\mathrm{M}[c] \ge 0$. By (38) and $\mathrm{M}[c] \ge 0$,

$$\mathrm{M}\big[\Lambda.(T_0)\big] - \gamma_k\,\mathrm{M}[c] \;\ge\; \mathrm{M}\big[\Lambda.(T_0)\big] - \Lambda_{\{k\}}(T_0)\,\mathrm{M}[c],$$

and by the recursion defining (25) the right side is the corresponding quantity for the set T_0 and the list obtained by prepending k to $z_1, \ldots, z_\ell$, which is nonnegative by the induction hypothesis. Dividing by $\mathbb{P}(E)$ gives (39).

Taking $\ell = 0$ and $v = k$ in (39) gives the transfer inequality

$$\mathbb{P}\big(H(R;\{k\}, Y \cup T_0)\big)\,\Lambda_{\{k\}}(T_0) \;\le\; \mathbb{P}\big(k \not\leftrightarrow Y \cup T_0\big)\,\Lambda_R(T_0) \tag{40}$$

for every k outside $Y \cup T_0 \cup R$. We now prove $\Lambda_R(T) \ge 0$ by induction on the cardinality of T. Let $k \in T$ have the largest value of r, put $T_0 = T \setminus \{k\}$, $E = \{k \not\leftrightarrow Y \cup T_0\}$ and $H_R = H(R;\{k\}, Y \cup T_0)$, so that (36) reads $\Lambda_R(T) = \Lambda_R(T_0) + \mathbb{E}[F(C_R) - m_k; H_R]$. On H_R we have $C_k \subseteq C_R$, so

$$\mathbb{E}\big[F(C_R) - m_k; H_R\big] \;\ge\; \mathbb{E}\big[F(C_k) - m_k; H_R\big].$$

Conditionally on the complete set of open pairs met by paths from k, and on E, the conditional probability of H_R is $\alpha_{R,Y\cup T_0}(C_k)$ with α as in (28) and $U = V$, an increasing function of that data by Lemma 7.9, and $\mathbb{E}[\alpha_{R,Y\cup T_0}(C_k); E] = \mathbb{P}(H_R)$ by (29). Hence (5) applied with $s = k$, with the set $Y \cup T_0$ and with the two increasing functions $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$ and $\mathcal{C} \mapsto \alpha_{R,Y\cup T_0}(\{k\} \cup \operatorname{span} \mathcal{C})$ gives $\mathbb{P}(E)\mathbb{E}[F(C_k); H_R] \ge \mathbb{P}(H_R)\mathbb{E}[F(C_k); E]$, whence

$$\mathbb{P}(E)\,\mathbb{E}\big[F(C_k) - m_k; H_R\big] \;\ge\; -\mathbb{P}(H_R)\,\gamma_k.$$

Multiplying the peeling identity by $\mathbb{P}(E)$ and using (40), (38) and the last display,

$$\mathbb{P}(E)\,\Lambda_R(T) \;\geq\; \mathbb{P}(H_R)\,\Lambda_{\{k\}}(T_0) - \mathbb{P}(H_R)\gamma_k \;\geq\; 0,$$

and dividing by $\mathbb{P}(E) > 0$ finishes the induction.

It remains to pass to weights in $[0,1]$. As in (17), compatibility of r with the numbers m_x gives

$$\Lambda_R(T) = \mathbb{E}\Big[\mathbf{1}\{R \nleftrightarrow Y,\, R \leftrightarrow T\}\Big(F(C_R) - \min_{x\in T\cap C_R} m_x\Big)\Big],$$

an expression not referring to r and continuous at every weight vector at which $\mathbb{P}(x \nleftrightarrow Y) > 0$ for each $x \in T$. Approximating such a vector by vectors with all coordinates strictly inside the unit interval, choosing a compatible injective r for each and passing to a subsequence along which the induced order on T is constant, the case already proved gives the assertion in the limit. $\square$

7.7 Proof of Theorem 7.1

Two identities relate F to the function Φ_a of (18) for a set of two vertices. Let $R = \{v, w\}$ and let a be a vertex.

Lemma 7.17. *For every family $\mathcal{S}$ of sets of pairs,*

$$\mathbb{E}\big[F(C_R) - F(C_a);\; R \nleftrightarrow a,\; \mathcal{E} \in \mathcal{S}\big] = \mathbb{E}\big[\Phi_a(C_R);\; R \nleftrightarrow a,\; \mathcal{E} \in \mathcal{S}\big],$$

where $\mathcal{E}$ denotes the complete set of open pairs met by paths from R. Furthermore, if $\mathbb{E}F(C_a) \leq \mathbb{E}F(C_v)$ then $\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a] \geq 0$.

Proof. For the identity, sum over the values of $\mathcal{E}$. On $\{R \nleftrightarrow a\}$ the vertex a lies outside C_R, every pair with exactly one endpoint in C_R is closed and the pairs with both endpoints outside C_R are fresh, so the conditional mean of $F(C_a)$ is $F(C_R) - \Phi_a(C_R)$.

For the inequality, assume $a \neq v$, since otherwise $\{R \nleftrightarrow a\}$ is empty by reflexivity. Let $\pi(K)$ be the probability that $w \nleftrightarrow a$ in a fresh configuration with every pair meeting $\{v\} \cup \operatorname{span} K$ deleted. Enlarging K deletes more pairs, so π is nonnegative and increasing, and summing over the values of $\mathcal{C}_v$ gives, for every real function p of sets of pairs,

$$\mathbb{E}\big[p(\mathcal{C}_v);\; R \nleftrightarrow a\big] = \mathbb{E}\big[p(\mathcal{C}_v)\pi(\mathcal{C}_v);\; v \nleftrightarrow a\big].$$

Apply (5) with $s = v$, with $X = \{a\}$ and with the increasing functions $p(\mathcal{C}) = \Phi_a(\{v\} \cup \operatorname{span} \mathcal{C})$ and π:

$$\mathbb{E}\big[\Phi_a(C_v);\; v \nleftrightarrow a\big]\,\mathbb{E}\big[\pi(\mathcal{C}_v);\; v \nleftrightarrow a\big] \;\leq\; \mathbb{P}(v \nleftrightarrow a)\,\mathbb{E}\big[\Phi_a(C_v);\; R \nleftrightarrow a\big].$$

The first factor on the left is $\mathbb{E}F(C_v) - \mathbb{E}F(C_a) \geq 0$ by Lemma 5.1(ii) and the second is nonnegative, so if $\mathbb{P}(v \nleftrightarrow a) > 0$ then $\mathbb{E}[\Phi_a(C_v); R \nleftrightarrow a] \geq 0$, while if $\mathbb{P}(v \nleftrightarrow a) = 0$ the event $\{R \nleftrightarrow a\}$ is null. Since Φ_a is increasing and $C_v \subseteq C_R$, the identity of the first part with $\mathcal{S}$ containing every set of pairs gives $\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a] = \mathbb{E}[\Phi_a(C_R); R \nleftrightarrow a] \geq \mathbb{E}[\Phi_a(C_v); R \nleftrightarrow a] \geq 0$. $\square$

Proof of Theorem 7.1. Put $R = \{v, w\}$ and $T = A \setminus \{a\}$. If $a = v$ or $a = w$ then $\{R \nleftrightarrow a\}$ is empty by reflexivity and both sides of (20) vanish. If $v \in T$ or $w \in T$ then $\{R \leftrightarrow T\}$ is the sure event, so the left side of (20) is $\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a]$; the hypothesis gives $\mathbb{E}F(C_a) \leq \mathbb{E}F(C_v)$ in the first case and $\mathbb{E}F(C_a) \leq \mathbb{E}F(C_w)$ in the second, and Lemma 7.17 applies, its two vertices being interchangeable. In the remaining case R is disjoint from A.

Let T_1 be the set of $x \in T$ with $\mathbb{P}(x \nleftrightarrow a) > 0$. Apply Theorem 7.16 with $Y = \{a\}$, with the set T_1, with the increasing function Φ_a and with the set R, which is disjoint from $\{a\} \cup T_1$. By Lemma 5.1(ii) the number (10) attached to $x \in T_1$ is $(\mathbb{E}F(C_x) - \mathbb{E}F(C_a))/\mathbb{P}(x \nleftrightarrow a) \geq 0$, so the subtracted sum is nonnegative and

$$\mathbb{E}\big[\Phi_a(C_R);\ R \nleftrightarrow a,\ R \leftrightarrow T_1\big] \ \geq\ 0.$$

The event $\{R \leftrightarrow T_1\}$ is determined by the complete set of open pairs met by paths from R, so the first part of Lemma 7.17 converts this into $\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a, R \leftrightarrow T_1] \geq 0$. Finally, a configuration in $\{R \nleftrightarrow a\} \cap \{R \leftrightarrow T\}$ but not in $\{R \leftrightarrow T_1\}$ has R joined to some $x \in T \setminus T_1$ while missing a, hence lies in the null event $\{x \nleftrightarrow a\}$; a finite union of null events is null, so the two set integrals agree and (20) follows. □

8 Deleting the edges at the starting vertex

Question 2.9 selects the vertex a by a probability computed in the graph H obtained by deleting every pair incident to o, and then asks for (4) in the original graph. Exposing the pairs at o turns the cluster of o into the union of the clusters of a random set of vertices of H, and Theorem 7.1 applies to each value of that set, once its conclusion is available for sets of every finite size.

Theorem 8.1. *Assume that* (20) *holds for every finite set R of vertices in place of a set of two. Let A be a nonempty finite set of vertices, let b be a vertex, let H be obtained from the given graph by deleting every pair incident to o and let $a \in A$ minimise $\mathbb{P}_H(x \leftrightarrow b)$ over $x \in A$. Then* (4) *holds. In particular Question 2.9 has an affirmative answer.*

Proof. Condition on the states of the pairs incident to o. For such a state η let N_η be the set of vertices $u \neq o$ with $\{o, u\}$ open in η, and put $R_\eta = \{o\} \cup N_\eta$. The pairs not incident to o have exactly the law of percolation on H, and they are independent of η. In the original graph a path leaving o starts with a pair of η and continues in H, possibly returning to o and leaving again, so, pointwise in the configuration on H,

$$C_o = C^H_{R_\eta}, \qquad \{o \leftrightarrow A\} = \{R_\eta \leftrightarrow_H A\}, \tag{41}$$

where the right sides are computed in H, in which o is isolated and therefore contributes the single vertex o to the union. Moreover, on $\{o \nleftrightarrow a\}$ no path from a can reach o, so no such path uses a pair of η, and C_a is the same in the two graphs.

Put $F = \mathbf{1}\{b \in \cdot\}$. On $\{o \leftrightarrow a\}$ the clusters of o and a coincide, so

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) - \mathbb{P}(a \leftrightarrow b, o \leftrightarrow A) = \mathbb{E}\big[F(C_o) - F(C_a);\ o \nleftrightarrow a,\ o \leftrightarrow A \setminus \{a\}\big],$$

and by (41) the right side equals

$$\sum_\eta \mathbb{P}(\eta)\, \mathbb{E}_H\big[F(C_{R_\eta}) - F(C_a);\ R_\eta \nleftrightarrow a,\ R_\eta \leftrightarrow A \setminus \{a\}\big].$$

The hypothesis on a says $\mathbb{E}_H F(C_a) \leq \mathbb{E}_H F(C_x)$ for every $x \in A$, so each summand is nonnegative by the assumed form of (20) in the graph H with the set R_η. Averaging gives (4). No division is used, so states η of probability zero and coincidences among o, a, b and the vertices of A are covered. □

9 A four-vertex example

Question 2.8 selects the vertex a by the probability of the joint event that a reaches b and that o does not reach A. When several vertices of A realise the minimum, the reading in which every one of them satisfies (4) is false.

Proposition 9.1. *There are a graph with four vertices, weights in* $[0,1]$*, a set* A *with two elements and vertices* o *and* b *outside* A *such that both elements of* A *minimise* $\mathbb{P}(x \leftrightarrow b, o \not\leftrightarrow A)$ *over* $x \in A$ *while* (4) *fails for one of them.*

Proof. Let $V = \{o, a_1, a_2, b\}$ and $A = \{a_1, a_2\}$, give the pairs $\{o, a_1\}$ and $\{a_2, b\}$ weight one and the pair $\{a_1, a_2\}$ weight one half, and give every other pair weight zero. Since $\{o, a_1\}$ is open with probability one and $a_1 \in A$, the event $\{o \leftrightarrow A\}$ has probability one, so $\mathbb{P}(x \leftrightarrow b, o \not\leftrightarrow A) = 0$ for both $x \in A$ and both vertices of A minimise. The event $\{a_2 \leftrightarrow b\}$ has probability one, so

$$\mathbb{P}(o \leftrightarrow A,\ a_2 \leftrightarrow b) = 1.$$

The event $\{o \leftrightarrow b\}$ requires the pair $\{a_1, a_2\}$ to be open, so $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) = \frac{1}{2}$. Thus (4) fails for $a = a_2$. □

The example is degenerate in the sense that the minimised quantity vanishes at both vertices of A. If instead a is required to be the unique minimiser, no counterexample is known and the question is open. If a is chosen to minimise $\mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$ rather than $\mathbb{P}(x \leftrightarrow b, o \not\leftrightarrow A)$, then (4) holds, by (1).

10 Verification status

The statements of this note have been checked to different depths, and this section records which.

The following are proved in Lean 4 with Mathlib, on top of the formalization of the conditioned covariance inequalities of Theorem 3.2. The toolchain is Lean `4.32.0` with Mathlib commit `81a5d257c8e410db227a6665ed08f64fea08e997`; each theorem listed compiles with no `sorry` and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

- Theorem 4.1: `KNAll.genY_of_surplusTransferY` in `KN/AvoidedGen.lean` for the induction, and `KNAll.genY_all` in `KN/Conjectures.lean` for the form quantified over all weights. Proposition 4.2: `KNAll.surplusMarginY_nonneg_of_csh` in `KN/AvoidedPeel.lean`, `KNAll.surplusTransferY_nondegenerate` in `KN/AvoidedTransfer.lean` and `KNAll.surplusTransferY_of_nondegenerate` in `KN/AvoidedClosure.lean`.
- Lemma 5.1: `KNAll.monotone_projFunA`, `KNAll.setIntegral_sub_eq_projFunA_conn` and `KNAll.setIntegral_projFunA_avoid` in `KN/Projection.lean`.
- Theorem 5.2: `KNAll.conjecture4Fixed_holds` in `KN/Conjectures.lean`.
- Corollary 5.4 and Lemma 5.3: `KNAll.conjecture4_holds` in `KN/Conjectures.lean` and `KNAll.conjecture4_clusterProperty_holds`, `KNAll.conjecture4_size` and `KNAll.conjecture4_sizeGE` in `KN/ClusterProperty.lean`.
- Corollary 5.5: `KNAll.conjecture2Strong_holds`, `KNAll.conjecture2_holds`, `KNAll.conjecture1_holds`, `KNAll.conjecture3_holds` and `KNAll.question7_holds` in `KN/Conjectures.lean`.

- Lemma 6.1 and Theorem 6.2: `KNAll.question5_dual` in `KN/Question5Dual.lean` and `KNAll.question5_holds` in `KN/Question5.lean`, the latter containing the separation argument.
- The reductions of Section 7.1: `KNAll.conjecture6_of_conjecture6Strong`, `KNAll.conjecture6_hypotheses_vacuous` and `KNAll.conjecture6Strong_of_forceOpen_min` in `KN/Conjecture6Reduction.lean`, the last giving (21) when the vertex a also minimises $\mathbb{P}^{e}(x \leftrightarrow b)$.
- Lemma 7.17: `KNAll.setIntegral_sub_eq_projFunA_pair_conn` and `KNAll.overlap_pair` in `KN/PairSource.lean`.

The remaining results of Section 7, namely Propositions 7.13, 7.14 and 7.15, Theorem 7.16, Theorem 7.1 and Theorem 7.4, together with the supporting Lemmas 7.5 to 7.12 and Lemmas 7.2 and 7.3, are proved on paper only. Their formalization is in progress; a skeleton containing 45 definitions and 59 theorem statements in the shape of those results compiles against the same toolchain.

Theorem 8.1, and with it the affirmative answer to Question 2.9, is proved on paper only, and it is stated conditionally on (20) for sets of every finite size, whereas Theorem 7.1 is proved for sets of at most two vertices.

Question 2.8 is open in the reading in which the vertex a is the unique minimiser. Proposition 9.1, which refutes the reading in which every minimiser is allowed, is an exact finite computation.

This note and the development it reports were produced entirely by AI systems, Claude Fable 5.1 (max), Claude Opus, and ChatGPT 5.6 sol (ultra) through Codex.

References

[1] R. Ahlswede and D. E. Daykin, *An inequality for the weights of two families of sets, their unions and intersections*, Z. Wahrscheinlichkeitstheorie verw. Gebiete **43** (1978), 183–185.

[2] J. van den Berg, O. Häggström and J. Kahn, *Some conditional correlation inequalities for percolation and related processes*, Random Structures Algorithms **29** (2006), 417–435. Theorem 1.1 on p. 3 is the first display of Theorem 3.1, and Theorem 1.4 on p. 7, with Remark 1 after Theorem 1.2 on p. 5 replacing the two vertices by two vertex sets, is the second.

[3] *Theta at the critical point vanishes for Bernoulli bond percolation on the hypercubic lattice in all dimensions at least two, in Lean 4 and Mathlib*, formal development, `https://github.com/anthropic-experimental/percolation`. Theorem 3.2 is its theorem `Percolation.Continuity.CSH.cshHolds`.

[4] T. E. Harris, *A lower bound for the critical probability in a certain percolation process*, Proc. Cambridge Philos. Soc. **56** (1960), 13–20.

[5] G. Kozma and S. Nitzan, *A reduction of the percolation at criticality problem to a conjectured inequality*, arXiv:2401.12397, 2024. Conjectures 1 and 2 and display (3) are on p. 3, Conjecture 3 on p. 15, Conjecture 4 and Question 5 on pp. 31–32, Conjecture 6 on p. 34, and Questions 7, 8 and 9 with display (41) on p. 36.